

Computer Netzwerke und verteilte Systeme Summer 2026



Exercise 5: Overlay Routing

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Deadline: The deadline for this exercise is **26.05.2026, 23:55** and in total **18 bonus points** can be achieved.

Bonus points: *Computer Netzwerke und verteilte Systeme* features a bonus point system to acknowledge first-rate performance. Exercise points count as bonus. Throughout the entire course, a bonus of up to a maximum of 0.7 can be obtained. The bonus does not help in passing the exam: You need to pass the exam with a 4.0 or better before any bonus is applied.

Group size: The recommended group size for submissions is 3. You may also submit your exercises as a group of 2 or alone if you do not find a group.

Important: Every theoretical task and subtask require proper sources. Submissions without proper sources will automatically receive 0 points.

For most of the task the slides of the lecture will be enough. For **research** task, use the materials provided in the lecture, or the internet.

Refrain from copying and pasting your answers, as this is considered plagiarism and will have **consequences**, as will using AI tools to create your answers. Instead, explain your answers in your own words and cite your sources!

Write down your answers in a legibly and unambiguously way, either by creating formatted documents (LaTeX, Word, typst, etc.) or by handwriting (**ensure that your handwriting is legible**).

Refrain from using colors that create less contrast between the text and the background. For marking tasks you may use blue and orange.

Task 1: Group Exercise: Distance Vector Routing

Initialize the final routing tables for each node (A, B, C, D) in the given network topology with the given connection costs. Analyze the consequences of a link failure between C and D and create the updated tables for three rounds based on the previous tables. Explain the problems that occur and propose a possible solution to this problem.

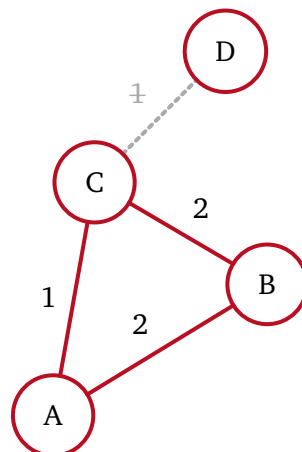


Figure 1: Topology for task 1. The dashed edge indicates the failed link between C and D.

Task 2: Group Exercise: Overlay Routing

State the routing paths in the overlay depicted in Figure 2, the hop count in the underlay, and the stretch factor for the link from A to E.

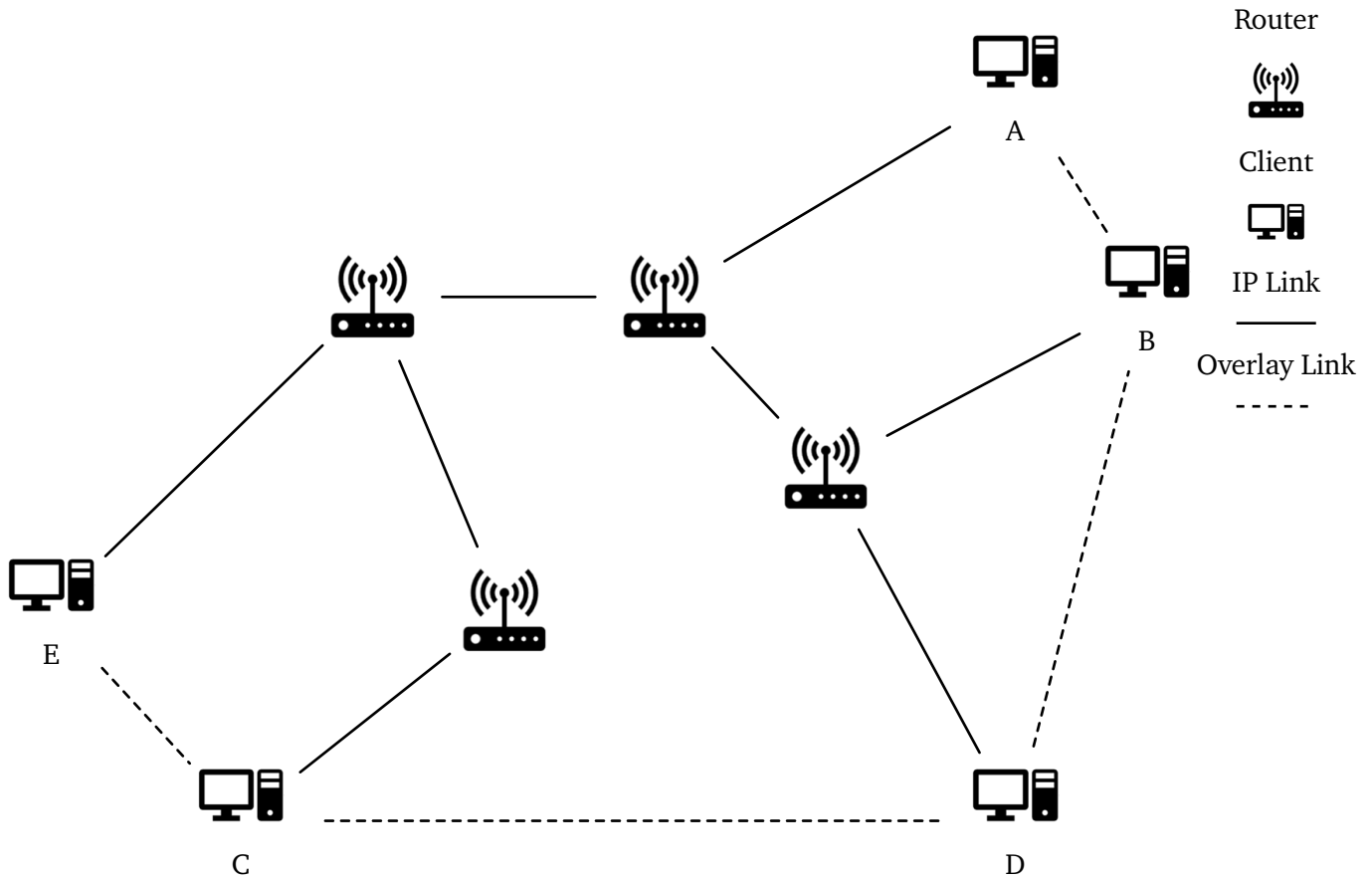


Figure 2: Overlay Routing Topology for Task 2

Task 3: Overlay Network Theory (6 Points)

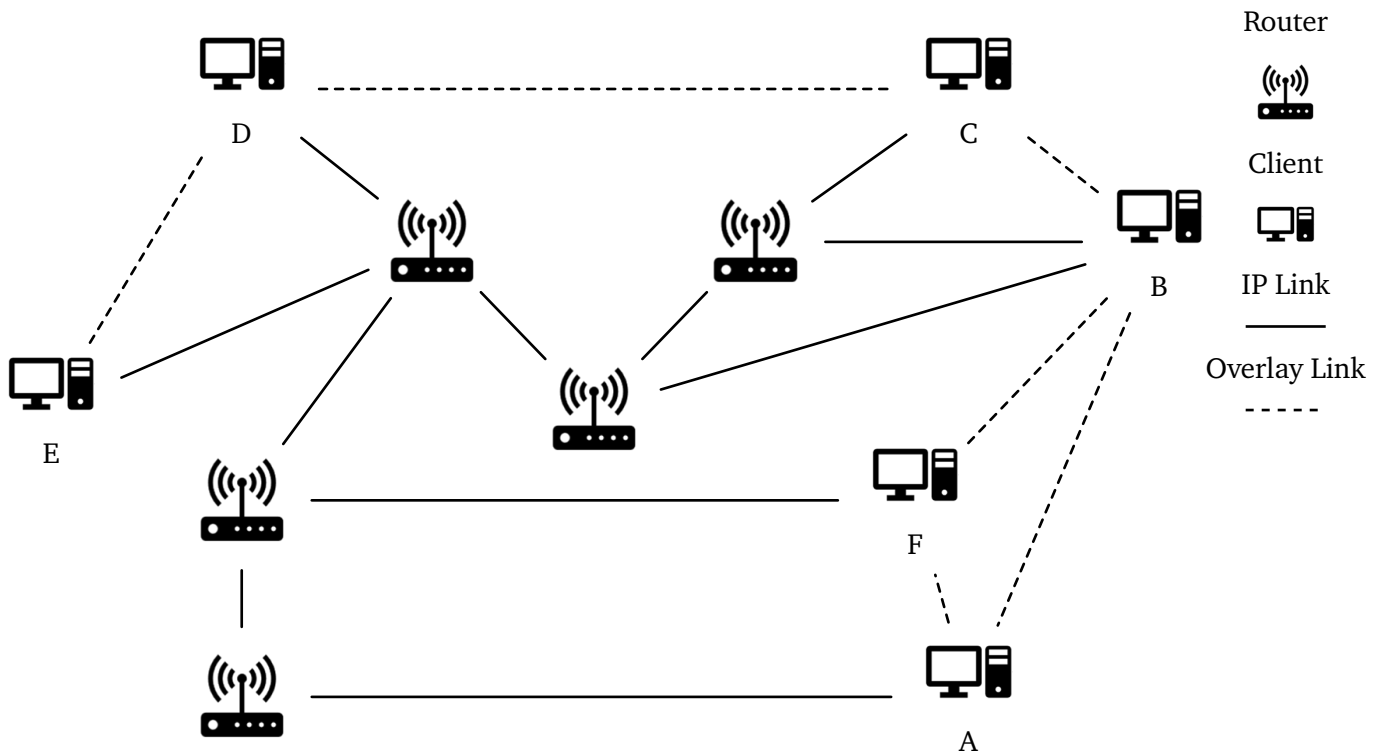
- (a) How does a “neighbor” in an overlay network differ from a “neighbor” in a traditional physical IP network (the underlying network)? (1 P)
- (b) Explain what happens when the Stretch is minimized. (1 P)
- (c) Explain the two important goals for overlay routing. Which method does each of them use to determine the best path? (2 P)
- (d) Name one reason why an overlay network is necessary for Peer-to-Peer (P2P) services and list two applications that employ overlay routing. (2 P)

Attention: Submissions without references will not be graded! Please add sources.

Task 4: Overlay Routing Practice (8 Points)

State the routing paths in the overlay depicted in Figure 3, the hop count in the underlay, and the stretch factor for the:

- (a) Link from *C* to *D*. (2 P)
- (b) Link from *B* to *D*. (2 P)
- (c) Link from *F* to *C*. (2 P)
- (d) Link from *E* to *F*. (2 P)



Task 5: Transport Layer Theory (4 Points)

- (a) Name the two primary transport-layer protocols of the Internet. Which services do they provide, and to which category of protocols do they belong? (2 P)
- (b) What is the Internet approach to identifying remote processes, and how does it work? (1 P)
- (c) Explain the difference between the responsibilities of the Network layer and the Transport layer. (1 P)

Attention: Submissions without references will not be graded! Please add sources.